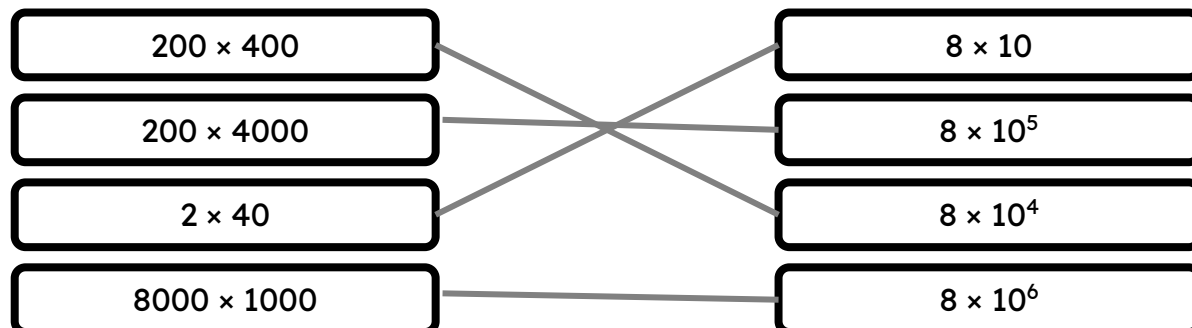




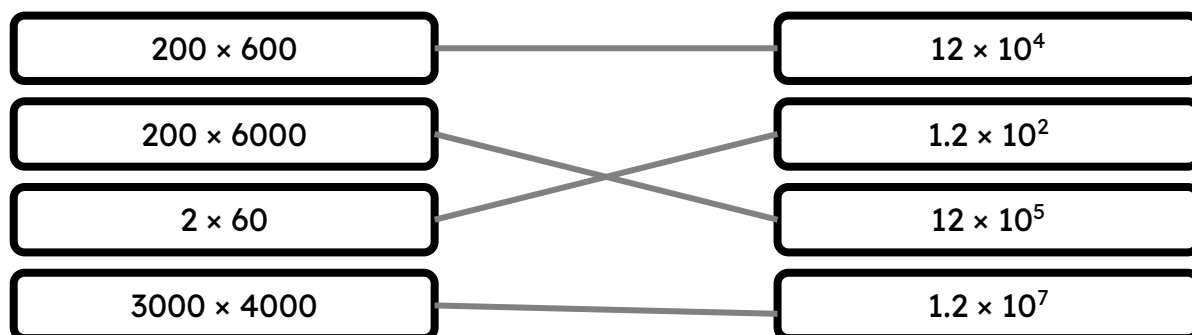
Changing compound interest rates

Task A: Using powers of 10 to simplify calculations

1) Match the calculation with the calculation using exponential form.



2) Match the calculation with the calculation using exponential form.



3) Fill in the missing numbers. Make sure to show your working out.

a) $9000 \times 100 = 9 \times 10^5$

$$\begin{aligned}
 &9 \times 1000 \times 1 \times 100 \\
 &9 \times 1 \times 1000 \times 100 \\
 &9 \times 10\,000 \\
 &9 \times 10^5
 \end{aligned}$$

b) $300 \times 250 = 75 \times 10^3 = 7.5 \times 10^4$

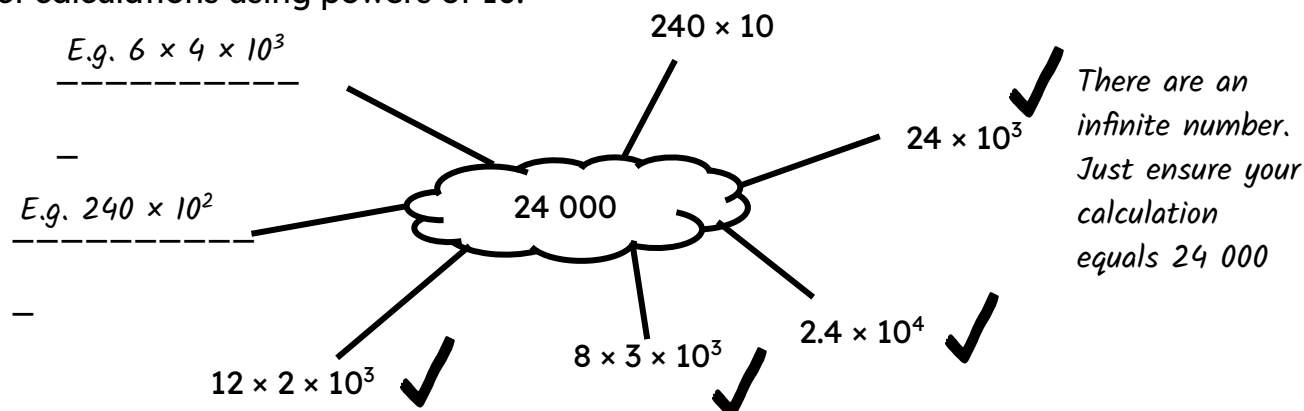
$$\begin{aligned}
 &3 \times 100 \times 25 \times 10 \\
 &3 \times 25 \times 100 \times 10 \\
 &75 \times 1000 \\
 &75 \times 10^3 \\
 &7.5 \times 10 \times 10^3 \\
 &7.5 \times 10 \times 10^3 \\
 &7.5 \times 10^4
 \end{aligned}$$

c) $4000 \times 1500 = 60 \times 10^5 = 6 \times 10^6$

$$\begin{aligned}
 &4 \times 1000 \times 15 \times 100 \\
 &4 \times 15 \times 1000 \times 100 \\
 &60 \times 100\,000 \\
 &60 \times 10^5 \\
 &6 \times 10 \times 10^5 \\
 &6 \times 10 \times 10^5 \\
 &6 \times 10^6
 \end{aligned}$$



4) Put a tick next to the calculations which are correct and create two of your own correct calculations using powers of 10.



TASK B: Using equivalent fractions to simplify calculations

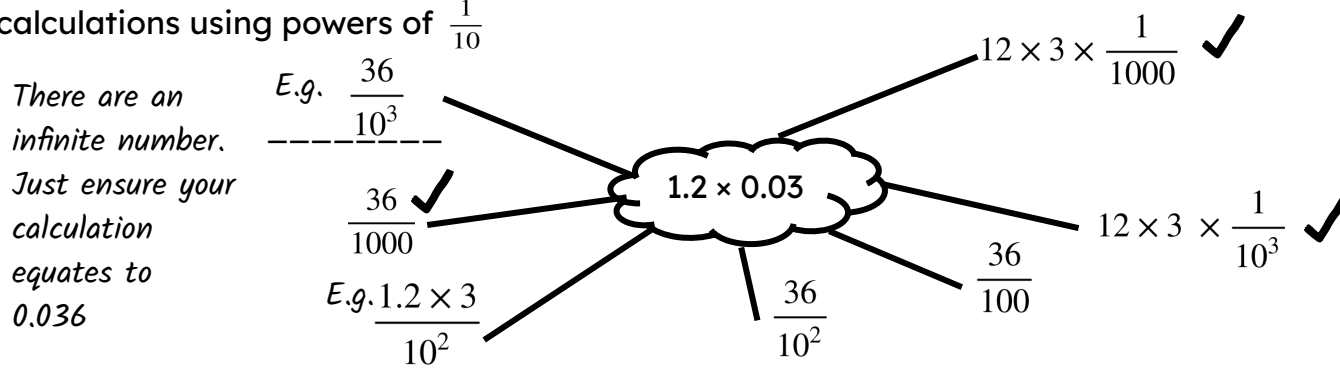
1) Identify the equivalent calculations to 3.4×9.3

- a** $34 \times 93 \times \frac{1}{100}$ ✓
- b** $34 \times 93 \times \frac{1}{10^2}$ ✓
- c** $\frac{34 \times 93}{100}$ ✓
- d** $\frac{34 \times 93}{1000}$
- e** $\frac{34 \times 93}{10^2}$ ✓

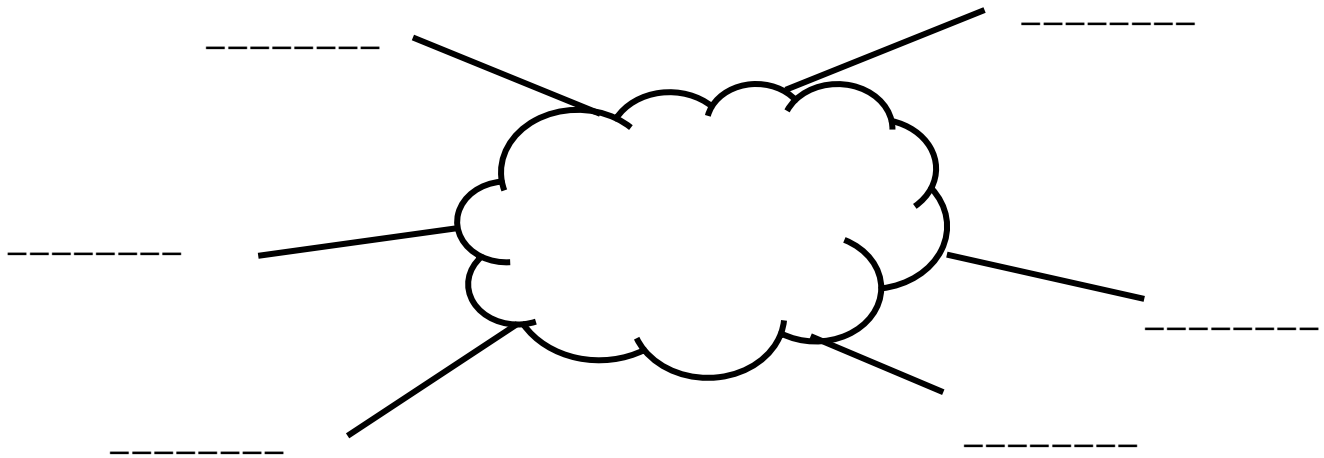
2) Identify the equivalent calculations to 0.4×3.8

- a** $4 \times 38 \times \frac{1}{100}$ ✓
- b** $4 \times 38 \times \frac{1}{1000}$
- c** $\frac{4 \times 38}{100}$ ✓
- d** $\frac{4 \times 38}{10^2}$ ✓
- e** $\frac{152}{100}$ ✓ This required you to calculate the answer to $4 \times 38 = 152$

3) Put a tick next to the calculations which are correct and create two of your own correct calculations using powers of $\frac{1}{10}$



4) Create your own calculation and six equivalent calculations using powers of $\frac{1}{10}$



There are an infinite number of answers. Ensure to check with a calculator or written methods.